

Emotional Assistance*

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Abstract—This paper presents a novel therapeutic conversation agent that integrates multimodal emotion recognition technologies with standardized mental health assessments to deliver personalized therapeutic interactions. The system combines facial emotion recognition through computer vision, speech emotion recognition through audio analysis, and standardized psychological questionnaires to create a comprehensive understanding of a user’s emotional and mental state. This information goes to the logic agent to make a system prompt appropriate for a therapeutic dialogue to a large language model. Initial evaluations show promising results in accurately detecting emotions and providing emotional support. This paper details the architecture, implementation and evaluation of this integrated system. This system has potential applications in supplementary mental health support, remote patient monitoring, and as an accessible first-line resource in regions with limited access to mental health professionals. The multimodal approach enables more nuanced and personalized support than text-only systems, potentially increasing user trust and adherence.

I. INTRODUCTION

Mental health conditions affect a significant portion of the global population. In 2019, 970 million people globally were living with a mental disorder, with anxiety and depression the most common. Besides, People with severe mental health conditions die 10 to 20 years earlier than the general population [1]. Despite the prevalence and impact of these conditions, access to mental health services remains limited due to factors including stigma, cost, and availability of qualified providers. Digital mental health interventions offer promising alternatives to traditional care models, potentially increasing accessibility and reducing barriers to treatment. Also, randomized controlled trials demonstrate that fully generative-AI therapy chatbots can significantly reduce clinical-level symptoms of anxiety and depression, with one RCT showing effect sizes comparable to low-intensity psychological interventions [2]. Emotion recognition plays a critical role in understanding user needs and enhancing communication. Facial Emotion Recognition (FER) refers to the process of identifying and categorizing human emotions based on facial expressions. By analyzing facial features and patterns, machines can learn about a person’s emotional state. This subfield of facial recognition is highly interdisciplinary, drawing on insights from computer vision, machine learning, and psychology [3]. Speech Emotion Recognition is a sub-discipline of affective computing that involves recognizing the emotional aspects of speech irrespective of the semantic content. A typical SER system can be considered as a collection of methodologies that isolate, extract and classify speech signals to detect

emotions embedded in them [4]. This project is about a system that integrates multiple emotion recognition modalities—facial expression analysis and speech emotion recognition—with standardized mental health assessments to inform a therapeutic conversation agent. The system aims to create a more empathetic and effective digital mental health intervention by combining these various inputs to gain a comprehensive understanding of the user’s emotional and mental state.

II. RELATED WORKS

A. Facial Emotion Recognition

Facial emotion recognition has been extensively studied in computer vision, with approaches ranging from geometric feature-based methods to deep learning models. Convolutional Neural Networks (CNNs) have demonstrated particularly strong performance on benchmark datasets such as FER2013 [5]. These models typically classify facial expressions into a discrete set of basic emotions, such as those proposed by Ekman [6]: anger, disgust, fear, happiness, sadness, and surprise, with neutrality often included as an additional category.

B. Speech Emotion Recognition

Early Speech Emotion Recognizer relied on spectral and prosodic features-MFCCs, ZCR, pitch, and energy—fed into classifiers like SVMs or GMMs. Recent trends emphasize deep representation learning, automatically extracting hierarchical features from raw audio or spectrograms, significantly improving robustness and generalization. Lightweight ensemble models combining CNNs and Bi-LSTMs have demonstrated competitive accuracy on benchmarks such as RAVDESS and TESS, especially when augmented with noise addition and spectrogram shifting. Comprehensive reviews also highlight challenges in class imbalance, cross-corpus generalization, and real-time deployment for low-resource devices[7].

C. Mental Health Questionnaires

Standardized mental health questionnaires are widely used in clinical settings for screening, diagnosis, and monitoring of mental health conditions. Common instruments include the Patient Health Questionnaire-9 (PHQ-9) [8] for depression, the Generalized Anxiety Disorder-7 (GAD-7) [9] for anxiety, the Perceived Stress Scale (PSS) for stress [10], and the WHO-5 Well-Being Index for overall psychological well-being [1]. These tools have been validated in numerous studies and provide valuable quantitative measures of mental health status.

D. LLM-based chatbot

Large language models (LLMs) have been rapidly adopted to power mental health chatbots, with over 50 systems identified—22 of which are explicitly LLM-based—addressing general distress, depression, anxiety, stress, and suicidality through conversational interfaces [11]. Randomized controlled trials (RCTs) demonstrate that expert-fine-tuned LLM chatbots can match low-intensity psychological interventions: an RCT of the “Therabot” system showed clinically significant reductions in anxiety and depressive symptoms over eight weeks (Cohen’s $d \approx 0.5\text{--}0.6$) [2], and a follow-up trial reported a mean symptom reduction for depression of 51% and 31% for generalized anxiety disorder after two months of use. A separate pilot study found that nearly 80% of participants reported ChatGPT helped them manage stress through mindfulness and relaxation exercises [12].

III. FACIAL EMOTION RECOGNITION

A. VGG-like with FER2013

1) *Introduction:* The goal is to develop a deep learning model that can accurately classify facial expressions into one of six categories: Angry, Fear, Happy, Sad, Surprise, and Neutral. I try a CNN model with a VGG-like structure to train the model with the FER2013 dataset. The directory in the repository where I did this model is Computer_vision/CNN.

2) *Data:* The FER-2013 dataset consists of 48x48 pixel grayscale images of faces that have been automatically registered to be centred and occupy a similar amount of space in each image. The dataset contains 24,400 images, with 22,968 examples in the training set and 1,432 examples in the public test set.

3) *Preprocessing:* I used data generators and data augmentation to train the deep learning model:

- Data generators create batches of images on-the-fly during training, making it possible to handle this large and complex dataset efficiently. Since loading all images into memory at once would be impractical, data generators allow the model to be trained on the entire dataset without exceeding memory limits.
- Data augmentation techniques to the training images, such as rotation, shifting, and flipping for increasing the size and diversity of the training set, which can improve the performance of the deep learning model. By introducing variability into the training process, data augmentation can also help to prevent overfitting, which occurs when the model becomes too closely tailored to the training set and performs poorly on new, unseen data.

4) *Training:* To develop the CNN, at first I defined the input layer and the number of filters in the first convolutional layer. Then, I added additional convolutional layers with increasing numbers of filters, followed by max-pooling layers to reduce the spatial dimensions of the feature maps. After the convolutional layers, I added fully connected layers with ReLU activation to classify the emotions.[13]

I experimented with different numbers of convolutional layers, filter sizes, and fully connected layers to optimize the model performance. Additionally, I used techniques such as dropout and batch normalization to prevent overfitting and improve the generalization capability of the model.

The structure that I finally chose is:

TABLE I
SUMMARY OF THE CNN ARCHITECTURE

Layer	Filters	Kernel Size	Activation
Input	-	$48 \times 48 \times 1$	-
Conv2D + BatchNorm	32	3×3	ReLU
Conv2D + BatchNorm	64	3×3	ReLU
MaxPooling2D + Dropout	-	2×2	-
Conv2D + BatchNorm	128	3×3	ReLU
Conv2D + BatchNorm	128	3×3	ReLU
MaxPooling2D + Dropout	-	2×2	-
Conv2D + BatchNorm	256	3×3	ReLU
Conv2D + BatchNorm	256	3×3	ReLU
MaxPooling2D + Dropout	-	2×2	-
Flatten	-	-	-
Dense + BatchNorm	256	-	ReLU
Dropout	-	-	0.5
Dense (Output)	7	-	Softmax

5) *Evaluation:* The training and validation loss function decreased over epochs with a final loss in the epoch 50 of loss: 0.9036 that it typically means the model is not performing optimally.

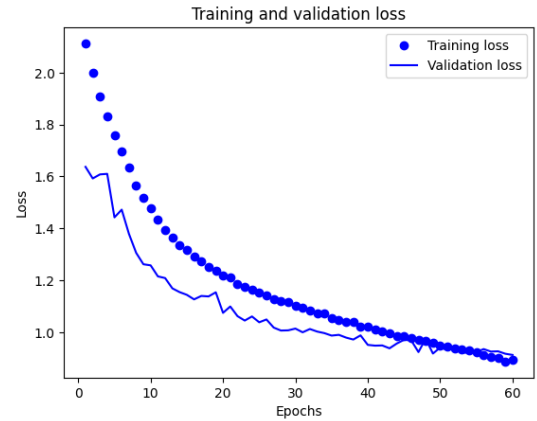


Fig. 1. Training and validation loss graph.

The training and validation accuracy increased over epochs with a final accuracy of 0.6631 that is not enough.

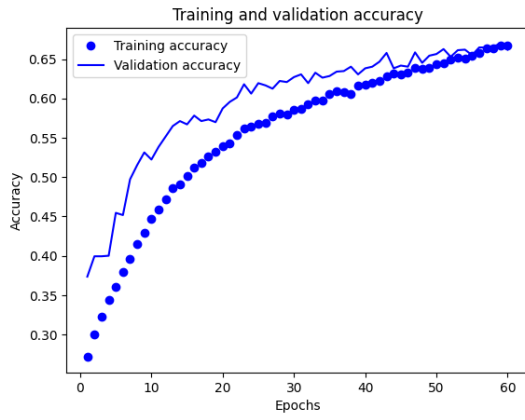


Fig. 2. Training and validation accuracy graph.

In the confusion matrix there are a lot of false positives and false negatives showing not enough accuracy. Besides, there are 0 disgust labeled images in the validation, this shows an imbalance in the data.

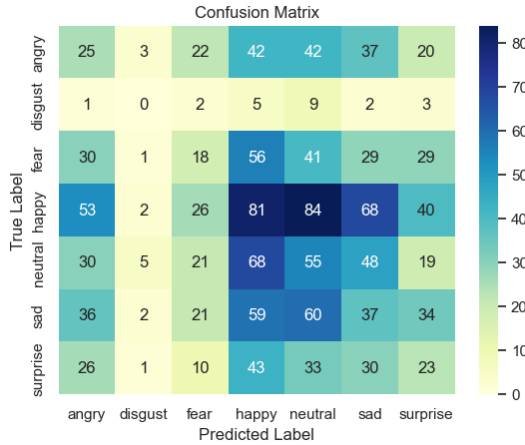


Fig. 3. Confusion matrix.

6) *Testing*: I used the `haarcascade_frontalface_default.xml` classifier from OpenCV, that is a widely used pre-trained model for detecting faces in images. Ergo, I recognized the faces with the haarcascade classifier and later recognise the facial emotion with the previous CNN. The haarcascade uses Haar features and is based on the Viola-Jones algorithm, which was one of the first methods to offer a real-time face detection solution. While it is quite efficient and works well for many applications, its performance can be limited by a complex environment, faces not in a frontal position and has lower accuracy compared to modern methods.

On the test that I did, the performance of the haarcascade was regular predicting sometimes false positives (it predicted as faces things that aren't a face) and false negatives above all when the face was not frontal. The performance of the VGG-like for facial emotion recognition wasn't good enough predicting the most of times neutral and happy when the true label were others.

B. emotion-ferplus-8

1) *Data*: I used the FER+ dataset is a notable extension of the original Facial Expression Recognition (FER) dataset (the one that I used previously). Developed to improve upon the limitations of the original dataset, FER+ offers a more refined and nuanced labeling of facial expressions. The FER+ has 8 emotions: happiness, sadness, anger, surprise, fear, disgust, neutral and contempt. The directory where I did this model is `Computer_vision/Custom_VGG13`.

2) *RFB-320 Single Shot Multibox Detector (SSD) Model for Face Detection*: Before the emotion is recognized, the face needs to be detected in the input frame like in the previous case. For this, the Ultra-lightweight Face Detection RFB-320 is used. It is a face detection model optimized for edge computing devices. Incorporating a modified Receptive Field Block (RFB) module (an extension of traditional convolutional layers), it effectively captures multiscale contextual information without adding computational overhead. Trained on the diverse WIDER FACE dataset (32,203 images and labeled 393,703 faces with a high degree of variability in scale, pose, and occlusion) and tailored for 320×240 input resolution, it boasts an impressive balance of efficiency with a computational rate of 0.2106 GFLOPs and a compact size housing of 0.3004 million parameters. Developed in the PyTorch framework, the model achieves a commendable mean average precision (mAP) of 84.78%, marking its position as a potent solution for efficient face detection in resource-constrained environments. In this implementation, the RFB-320 SSD (object detection algorithm that detects and classifies objects in an image in a single forward pass of the network) model has been used in Caffe format (the data storage and model representation format used by Caffe, a deep learning framework). [3]

3) *Custom VGG13 Model Architecture*: The emotion recognition classification model employs a customized VGG13 architecture designed for 64×64 grayscale images. It classifies images into eight emotion classes using convolutional layers with max pooling and dropout to prevent overfitting. This enhances the model's ability to generalize and accurately recognize emotions in various images. It was training using the FER+ dataset. The structure is:

TABLE II
CUSTOMIZED VGG13 MODEL ARCHITECTURE FOR FACIAL EMOTION
RECOGNITION

Layer	Filters	Kernel Size	Activation
Input	-	$64 \times 64 \times 1$	-
Conv2D	64	3×3	ReLU
Conv2D	64	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Conv2D	128	3×3	ReLU
Conv2D	128	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
Conv2D	256	3×3	ReLU
MaxPooling2D	-	2×2	-
Dropout	-	-	0.25
Fully Connected (Dense)	1024	-	ReLU
Dropout	-	-	0.50
Fully Connected (Dense)	1024	-	ReLU
Dropout	-	-	0.50
Fully Connected (Dense)	8	-	ReLU
Softmax (Output)	8	-	Softmax

4) *Implementation*: I employed the emotion-ferplus-8.onnx model, that is a pretrained model of learnopencv, for real-time facial emotion recognition. The model was integrated with a live video stream captured via webcam. Face detection was performed using the SSD-based RFB-320 detector with a Caffe backend. Preprocessing steps included image normalization using a mean of [127, 127, 127] and standard deviation of 128.0, followed by the computation of prior boxes based on feature maps derived from multiple strides ([8, 16, 32, 64]). The detection process leveraged bounding box decoding, confidence thresholding, and Non-Maximum Suppression (NMS) with an IoU threshold of 0.3 to filter predictions. The detected face region was then passed to the ONNX model to classify into one of seven emotion categories: neutral, happiness, surprise, sadness, anger, disgust, and fear. Output video was optionally recorded for subsequent analysis.

5) *Testing*: The ultralightweight face detection RFB320 is very good and accurate recognizing the faces although the quantity, angles of the face, illumination... The custom VGG13 model architecture performed well in classifying emotions from detected faces. It excelled at recognizing happiness, neutrality, and surprise, while showing slightly lower accuracy for sadness, disgust, and fear.

IV. SPEECH EMOTION RECOGNITION

This part is added in this directory of the repository: NLP/SER.

A. Data

- **Ravdess**: This portion of the RAVDESS contains 1440 files: 60 trials per actor x 24 actors = 1440. The

RAVDESS contains 24 professional actors (12 female, 12 male), vocalizing two lexically-matched statements in a neutral North American accent. Speech emotions includes calm, happy, sad, angry, fearful, surprise, and disgust expressions. Each expression is produced at two levels of emotional intensity (normal, strong), with an additional neutral expression [14].

- **Crema-D**: a data set of 7,442 original clips from 91 actors. These clips were from 48 male and 43 female actors between the ages of 20 and 74 coming from a variety of races and ethnicities (African America, Asian, Caucasian, Hispanic, and Unspecified). Actors spoke from a selection of 12 sentences. The sentences were presented using one of six different emotions (Anger, Disgust, Fear, Happy, Neutral, and Sad) and four different emotion levels (Low, Medium, High, and Unspecified) [15].
- **Tess**: There are a set of 200 target words were spoken in the carrier phrase "Say the word ." by two actresses (aged 26 and 64 years) and recordings were made of the set portraying each of seven emotions (anger, disgust, fear, happiness, pleasant surprise, sadness, and neutral). There are 2800 data points (audio files) in total [16].
- **Savee**: The SAVEE database was recorded from four native English male speakers (identified as DC, JE, JK, KL) aged from 27 to 31 years. Emotion has been described psychologically in discrete categories: anger, disgust, fear, happiness, sadness and surprise. This is supported by the cross-cultural studies of Ekman and studies of automatic emotion recognition tended to focus on recognizing these. We added neutral to provide recordings of 7 emotion categories. The text material consisted of 15 TIMIT sentences per emotion: 3 common, 2 emotion-specific and 10 generic sentences that were different for each emotion and phonetically-balanced. The 3 common and $2 \times 6 = 12$ emotion-specific sentences were recorded as neutral to give 30 neutral sentences [17].

B. Data augmentation

Data augmentation is the process by which we create new synthetic data samples by adding small perturbations on our initial training set. The objective is to make my model invariant to the perturbations and to enhance its ability to generalize.

- **Noise Injection**: Random Gaussian noise was added to the audio signals with amplitude proportional to the maximum amplitude of the original signal (noise factor of 0.035).
- **Time Stretching**: Audio signals were time-stretched using librosa's phase vocoder with a rate of 0.8, altering the duration without affecting pitch.
- **Time Shifting**: Random circular shifting of the audio signal within a range of ± 5000 samples was applied to simulate variations in emotion onset timing.
- **Pitch Shifting**: Pitch modification was implemented using librosa's pitch shift function with a factor of 0.7, altering the tonal characteristics while preserving temporal structure.

Through these augmentation techniques, the original dataset was effectively tripled in size, contributing significantly to the model's ability to generalize across diverse acoustic conditions.

C. Feature Extraction

I extracted a comprehensive set of acoustic features known to be relevant for emotion recognition:

- **Zero Crossing Rate (ZCR):** Measures the rate at which the signal changes from positive to negative, capturing speech rhythmic characteristics.
- **Chromagram:** Represents the spectral energy across the 12 pitch classes, capturing tonal content relevant to emotional expression.
- **Mel-Frequency Cepstral Coefficients (MFCCs):** Captures the short-term power spectrum of the sound, providing information about the vocal tract configuration that shapes different emotional expressions.
- **Root Mean Square Energy:** Represents the amplitude envelope of the signal, capturing intensity variations associated with emotional arousal.
- **Mel Spectrogram:** Provides a time-frequency representation adapted to human auditory perception, capturing both spectral and temporal dynamics of emotional speech.

Feature extraction was performed on each audio file with a duration of 2.5 seconds, offset by 0.6 seconds to eliminate non-speech segments at the beginning of recordings. The feature vectors extracted from original and augmented samples were combined into a comprehensive feature matrix for model training.

D. Model Architecture

The SER model employs a 1D Convolutional Neural Network (CNN) architecture designed to capture the temporal and spectral patterns characteristic of emotional speech:

TABLE III
CNN ARCHITECTURE FOR SPEECH EMOTION RECOGNITION

Layer	Units	Size	Parameters
Conv1D	256	5	stride=1, pad=same, ReLU
MaxPool1D	-	5	stride=2, pad=same
Conv1D	256	5	stride=1, pad=same, ReLU
MaxPool1D	-	5	stride=2, pad=same
Conv1D	128	5	stride=1, pad=same, ReLU
MaxPool1D	-	5	stride=2, pad=same
Dropout	-	-	rate=0.2
Conv1D	64	5	stride=1, pad=same, ReLU
MaxPool1D	-	5	stride=2, pad=same
Flatten	-	-	-
Dense	32	-	ReLU
Dropout	-	-	rate=0.3
Dense (Output)	8	-	Softmax

The model was compiled using the Adam optimizer with categorical cross-entropy loss and accuracy as the evaluation

metric. To prevent overfitting and improve convergence, we implemented learning rate reduction on plateau, decreasing the learning rate by a factor of 0.4 when the loss plateaued for two consecutive epochs.

E. Training and Evaluation

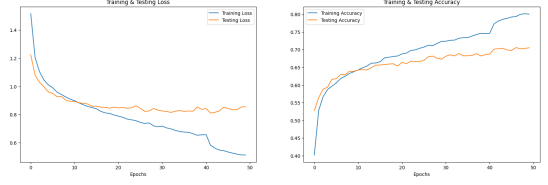


Fig. 4. Training and validation loss/accuracy curves for SER model.

The model has achieved an accuracy of 70.51% showing good performance but not perfect and a loss function of approximately 0.5%.

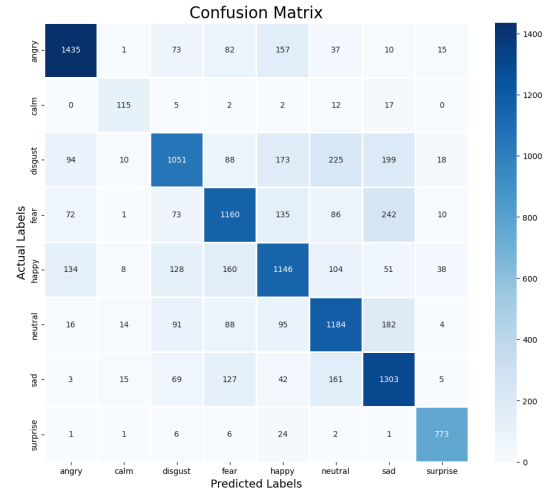


Fig. 5. Confusion matrix for SER model.

Analyzing the confusion matrix, model is more accurate in predicting surprise and angry emotions with less number of false negatives.

V. MENTAL HEALTH QUESTIONNAIRE

The mental health questionnaire is an important component of the emotional assistance designed to collect standardized psychological assessment data to complement the multimodal emotion recognition capabilities to be passed to a chatbot as a system prompt. This module administers validated clinical instruments to evaluate the user's psychological state across multiple dimensions, providing a more comprehensive understanding of their mental wellbeing beyond what can be detected through facial expressions and voice analysis alone.

A. Implemented Assessments

The questionnaire module incorporates four standardized clinical assessments, each targeting a different aspect of mental health:

- **Mood Assessment (PHQ-9 Modified):** A 9-item questionnaire evaluating symptoms of depression over the past two weeks. Questions cover key depression indicators including anhedonia, depressed mood, sleep disturbance, fatigue, appetite changes, feelings of worthlessness, concentration difficulties, psychomotor abnormalities, and suicidal ideation. Responses range from "Not at all" to "Nearly every day" with corresponding scores of 0-3.
- **Anxiety Assessment (GAD-7 Modified):** A 7-item questionnaire measuring anxiety symptoms over the past two weeks. Questions evaluate nervousness, worry control, excessive worry, relaxation difficulties, restlessness, irritability, and fear. The scoring system matches the PHQ-9 with responses from "Not at all" to "Nearly every day" (0-3 points).
- **Stress Assessment (PSS-10 Modified):** A 10-item questionnaire measuring perceived stress over the past month. This instrument evaluates both negative elements (feeling upset, unable to control important things, nervous) and positive coping factors (confidence in handling problems, feeling on top of things). The assessment uses a 5-point scale from "Never" to "Very often" (0-4 points), with reverse scoring applied to positive items.
- **Well-being Assessment (WHO-5 Modified):** A 5-item questionnaire measuring current psychological well-being. Questions assess positive mood, calmness, vitality, restfulness, and interest in daily activities. Responses range from "At no time" to "All the time" on a 6-point scale (0-5 points).

Each questionnaire generates a numerical score corresponding to validated clinical thresholds for symptom severity, allowing for meaningful interpretation and comparison over time.

B. Implementation

The questionnaire module was implemented using Streamlit, providing an intuitive interface for users to complete assessments either individually or as a comprehensive battery. Also, the scores are calculated automatically in based of the previous scales and mapped to clinical significance thresholds with corresponding interpretations (e.g., "Minimal," "Mild," "Moderate," "Severe"). Then, the result will be passed to the logic agent.

VI. LOGIC AGENT

The Logic Agent serves to generate the system prompt in base of the result of the assessment and the emotion recognized of the user. The functions of the Logic Agent are:

- **Multimodal Integration:** The agent reconciles potentially conflicting emotional signals from facial expressions and voice tone to determine the user's emotional state. It selects the final emotion with basic logic rules. For example, in cases where one label is 'neutral' and the other corresponds to a different emotion, the non-neutral label is selected. For the cases that both are contradictory, the output is neutral emotion.
- **Context Awareness:** The system maintains conversation history and assessment results to provide responses that acknowledge the user's evolving emotional and psychological state.
- **Prompt Engineering:** The system constructs specialized prompts for the language model that incorporate emotional context, assessment results, and appropriate therapeutic frameworks.

VII. CHATBOT

The chatbot is implemented with a system prompt that put the context of the mind state and emotion recognized of the user. For the implementation I tried with 3 different types of models with increasing level of complexity and capability. For the evaluation, I did 10 prompts and evaluated the response of each by my own criteria and did the average. The 10 prompts have different topic and emotions such as feeling overwhelmed with everything, being happy, having anxiety...

A. Naïve Bayes

My first approach was the Naïve Bayes classifier with a Bag-of-Words approach to map user inputs to appropriate therapeutic responses. I created pairs of training dataset of example emotional statements and appropriate empathetic responses, with a total of 42 examples. This approach computationally efficient and easily interpretable, this method was limited in generating contextual responses beyond its training examples. Besides, it does not store the user's prompt so it is impossible to handle a conversation. Finally, in the evaluation with the 10 prompts I put an average of 0,5.

B. BERT-like encoder

For the BERT-like encoder I tried with a pre-trained fine-tuned DistilRoBERTa-base for Emotion classification [18]. DistilRoBERTa-base is a transformer model that performs sentiment analysis. I tried with a fine-tuned model on transcripts from the Friends show with the goal of classifying emotions from text data, specifically dialogue from Netflix shows or movies. The model predicts 6 Ekman emotions and a neutral class. These emotions include anger, disgust, fear, joy, neutrality, sadness, and surprise. The model is a fine-tuned version of Emotion English DistilRoBERTa-base and DistilRoBERTa-base. This model was initially trained on:

- Crowdfower (2016)
- Emotion Dataset, Elvis et al. (2018)
- GoEmotions, Demszky et al. (2020)
- ISEAR, Vikash (2018)
- MELD, Poria et al. (2019)
- SemEval-2018, EI-reg, Mohammad et al. (2018)

This approach classifies very good the emotion of the prompts with a success of the 90%. However, it still relies on template-based responses rather than generating fully contextual conversation, which limits its therapeutic capabilities for complex interactions. Finally, it get a score of 1,3 because they were too general and were based only in the emotion.

C. LLM Decoder

My last type of model was a large language model with a decoder architecture, which offers significant advantages over encoder-based models for conversational applications because they are trained to predict the next token based on all previous tokens, making them perfect for generative tasks like conversation. I experimented with this three models: dialogpt-therapy-lora, llama-3.2-1b-instruct and deepseek-r1-distill-qwen-7b.

1) *dialogpt-therapy-lora*: I did a fine-tuned of the dialogpt-medium using LoRA (Low-Rank Adaptation) on the dataset: "LuangMV97/Empathetic_counseling_Dataset" [19]. This dataset contains 38,673 dialogue pairs in two columns: input, representing the user's utterance, and label, the expected empathetic response. LoRA was chosen due to its parameter-efficient fine-tuning capability, which significantly reduces the number of trainable parameters by freezing the original model weights and introducing small, trainable low-rank matrices into selected layers. This approach preserves the general knowledge of the pretrained model while allowing effective task-specific adaptation. The resulting model showed modest performance, with an evaluation score of 4.19, indicating potential areas for further improvement.

2) *llama-3.2-1b-instruct*: The llama model achieved the highest performance score (7.8), demonstrating strong empathetic, being coherent, emotionally resonant, and aligned well with therapeutic dialogue norms.

3) *deepseek-r1-distill-qwen-7b*: The deepseek model received the lowest score (2.0). Although it produced grammatically correct and elaborate responses, its performance was affected by verbosity and a lack of direct engagement with user inputs.

Finally, I chose the llama-3.2-1b-instruct model, even at smaller scales, it demonstrated strong performance and provided psychologically supportive.

VIII. APPLICATION INTERFACE

The user interface serves to combine the mental health assessment, the multi-modal emotion recognition and the chatbot with the user interaction made with streamlit. This application follows a tab-based layout that guides users to the 3 steps:

- **Emotion Detection Tab**: Captures and analyzes facial expressions and voice to determine emotional state. Finally with the VGG-like model because had a good performance with an accuracy of 70.51% and, with the Haar Cascade for frontal face of OpenCV library, has an easy implementation. The Haar cascade model was trained for detecting frontal human face and is perfect for the project because the user is supposed to be in front of the camera talking to the chatbot.
- **Mental Health Assessment Tab**: Administers standardized psychological questionnaires where user could select the topic and do all he wants.
- **Conversation Tab**: Facilitates therapeutic dialogue with an emotionally-aware agent. Users can interact with the chatbot either through spoken or written input. The

chatbot provides responses in both written form and synthesized audio. For speech-to-text conversion of the user's input, I employed Whisper, while for converting the chatbot's responses from text to speech, I used gTTS.

IX. CONCLUSION

This paper presents a novel therapeutic conversation agent that integrates multimodal emotion recognition with standardized mental health assessments to deliver personalized therapeutic interactions. The system demonstrates several key innovations:

A. Key Contributions

- **Multimodal Emotion Recognition Integration**: I successfully combined facial emotion recognition (finally using the VGG-like model) and speech emotion recognition (using a 1D CNN with 70.51% accuracy) to create a more comprehensive understanding of users' emotional states.
- **Standardized Assessment Integration**: The incorporation of validated clinical instruments (PHQ-9, GAD-7, PSS-10, and WHO-5) provides quantitative measures of psychological well-being that complement the emotion recognition capabilities.
- **Logic Agent Framework**: Our logic agent effectively reconciles potentially conflicting emotional signals from different modalities and integrates assessment results to construct contextually appropriate prompts for the language model.
- **Model Comparison**: Through systematic evaluation of different approaches (Naïve Bayes, BERT-like encoder, and various LLM decoders), we determined that the llama-3.2-1b-instruct model provided the most effective therapeutic responses (scoring 7.8 on our evaluation metrics) despite its relatively small size.

B. Limitations

Despite promising results, several limitations should be acknowledged:

- **Emotion Recognition Accuracy**: While the facial emotion recognition system performed well for certain emotions (happiness, neutrality, and surprise), it demonstrated lower accuracy for others (sadness, disgust, and fear). Similarly, the speech emotion recognition achieved 70.51% accuracy, indicating room for improvement.
- **Generalizability**: The models were trained on specific datasets (FER2013, RAVDESS, CREMA-D, TESS, SAVEE) which may not fully represent the diversity of human emotional expressions across different cultural contexts and demographics.
- **Real-world Validation**: While individual components were evaluated in controlled settings, comprehensive testing in real-world therapeutic scenarios with diverse user populations is needed to validate the system's clinical utility.
- **Privacy and Security**: The current implementation does not address potential privacy concerns related to processing sensitive emotional and mental health data.

C. Future Work

Several promising directions for future research emerge from this work:

- **Improve Emotion Recognition:** Incorporating additional modalities such as text sentiment analysis and physiological signals (heart rate variability, non-facial body emotion recognition) could further improve the system's ability to accurately detect emotional states.
- **Longitudinal Tracking:** Developing capabilities to monitor changes in emotional patterns and mental health metrics over time could enhance the therapeutic value of the system and provide valuable insights for users and clinicians.
- **Personalization:** Implementing adaptive learning mechanisms that tailor responses based on individual user preferences, response patterns, and therapeutic outcomes.
- **Clinical Validation:** Conducting randomized controlled trials to evaluate the system's effectiveness compared to traditional therapeutic approaches and other digital mental health interventions.
- **Cross-cultural Adaptation:** Expanding the training datasets and response frameworks to better accommodate cultural differences in emotional expression and therapeutic needs.

In conclusion, this multimodal emotional assistance system represents a promising step toward more empathetic and effective digital mental health interventions. By combining facial and speech emotion recognition with standardized psychological assessments and leveraging advanced language models, the system offers a more comprehensive approach to digital emotional support than text-only solutions. While not intended to replace professional mental health services, such systems could serve as valuable supplementary resources, particularly in regions with limited access to mental health professionals or as an initial engagement tool to reduce barriers to seeking help.

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